

A Practical Guide to AlphaGo

TOPIC -- ALPHAGO

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Similar systems Facebook has also been working on its own Go-playing system darkforest, also based on combining machine learning and Monte Carlo tree search. Although a strong player against other computer Go programs, as of early 2016, it had not yet defeated a professional human player. Darkforest has lost to CrazyStone and Zen and is estimated to be of similar strength to them. DeepZenGo, a system developed with support from video-sharing website Dwango and the University of Tokyo, lost 2–1 in November 2016 to Go master Cho Chikun, who holds the record for the largest number of Go title wins in Japan. A 2018 paper in Nature cited AlphaGo's approach as the basis for a new means of computing potential pharmaceutical drug molecules. Systems consisting of Monte Carlo tree search guided by neural networks have since been explored for a wide array of applications.

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AlphaGo's team published an article in the journal Nature on 19 October 2017, introducing AlphaGo Zero, a version without human data and stronger than any previous human-champion-defeating version. By playing games against itself, AlphaGo Zero surpassed the strength of AlphaGo Lee in three days by winning 100 games to 0, reached the level of AlphaGo Master in 21 days, and exceeded all the old versions in 40 days. In a paper released on arXiv on 5 December 2017, DeepMind claimed that it generalized AlphaGo Zero's approach into a single AlphaZero algorithm, which achieved within 24 hours a superhuman level of play in the games of chess, shogi, and Go by defeating world-champion programs, Stockfish, Elmo, and 3-day version of AlphaGo Zero in each case.

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AlphaGo played South Korean professional Go player Lee Sedol, ranked 9-dan, one of the best players at Go, with five games taking place at the Four Seasons Hotel in Seoul, South Korea on 9, 10, 12, 13, and 15 March 2016, which were video-streamed live. Out of five games, AlphaGo won four games and Lee won the fourth game which made him recorded as the only human player who beat AlphaGo in all of its 74 official games. AlphaGo ran on Google's cloud computing with its servers located in the United States. The match used Chinese rules with a 7.5-point komi, and each side had two hours of thinking time plus three 60-second byoyomi periods. The version of AlphaGo playing against Lee used a

similar amount of computing power as was used in the Fan Hui match. The Economist reported that it used 1,920 CPUs and 280 GPUs. At the time of play, Lee Sedol had the second-highest number of Go international championship victories in the world after South Korean player Lee Chang-ho who kept the world championship title for 16 years. Since there is no single official method of ranking in international Go, the rankings may vary among the sources. While he was ranked top sometimes, some sources ranked Lee Sedol as the fourth-best player in the world at the time. AlphaGo was not specifically trained to face Lee nor was designed to compete with any specific human players. The first three games were won by AlphaGo following resignations by Lee. However, Lee beat AlphaGo in the fourth game, winning by resignation at move 180. AlphaGo then continued to achieve a fourth win, winning the fifth game by resignation. The prize was US\$1 million. Since AlphaGo won four out of five and thus the series, the prize will be donated to charities, including UNICEF. Lee Sedol received \$150,000 for participating in all five games and an additional \$20,000 for his win in Game 4. In June 2016, at a presentation held at a university in the Netherlands, Aja Huang, one of the Deep Mind team, revealed that they had patched the logical weakness that occurred during the 4th game of the match between AlphaGo and Lee, and that after move 78 (which was dubbed the "divine move" by many professionals), it would play as intended and maintain Black's advantage. Before move 78, AlphaGo was leading throughout the game, but Lee's move caused the program's computing powers to be diverted and confused. Huang explained that AlphaGo's policy network of finding the most accurate move order and continuation did not precisely guide AlphaGo to make the correct continuation after move 78, since its value network did not determine Lee's 78th move as being the most likely, and therefore when the move was made AlphaGo could not make the right adjustment to the logical continuation.

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Technology and AI-related fields James Vincent, a reporter from The Verge, comments that "It prods and pokes viewers with unsubtle emotional cues, like a reality TV show would. "Now, you should be nervous; now you should feel relieved". The AlphaGo footage slowly captures the moment when Lee Sedol acknowledges the true power of AlphaGo AI. In the first game, he had more experience than his human-programmed AI, so he thought it would be easy to beat the AI. However, the early game dynamics were not what he expected. After losing the first match, he became more nervous and lost confidence. Afterward, he reacted to attacks by saying that he just wanted to win the match, unintentionally displaying his anger, and acting in an unusual way. Also, he spends 12 minutes on one move, while AlphaGo only takes a minute and a half to respond. AlphaGo weighs each alternative equally and consistently. No

reaction to Lee's fight. Instead, the game continues as if he was not there. James also said that "suffice to say that humanity does land at least one blow on the machines, through Lee's so-called "divine move". "More likely, the forces of automation we'll face will be impersonal and incomprehensible. They'll come in the form of star ratings we can't object to, and algorithms we can't fully understand. Dealing with the problems of AI will take a perspective that looks beyond individual battles. AlphaGo is worth seeing because it raises these questions" (Vincent, 2017) Murray Shanahan, a professor of cognitive robotics at Imperial College London, critics that "Go is an extraordinary game but it represents what we can do with AI in all kinds of other spheres," says Murray Shanahan, professor of cognitive robotics at Imperial College London and senior research scientist at DeepMind, says. "In just the same way there are all kinds of realms of possibility within Go that have not been discovered, we could never have imagined the potential for discovering drugs and other materials."